Homework 7

Question 1

The exact equation for the $\frac{d\bar{P}}{d\Omega}$ is given by

$$\frac{dP}{d\Omega} = \frac{1}{4\pi}\frac{1}{c^3}\left[\hat{r} \times \int d^3 r \, \frac{\partial \vec{j}(r', t_r)}{\partial t}\right]^2 \quad -----(1)$$

where $t_r = t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}$

Equation 10.233 from the notes was

$$\vec{j} = \begin{pmatrix} 0 \\ 0 \\ j_z \end{pmatrix} \quad \text{here } j_z = I \sin\left(\frac{kl}{2} - k \left| z \right|\right) \delta(x)\, \delta(y) \sin(\omega t)$$

Here $\omega = kc$ and $\left| z \right| \leq \frac{l}{2}$

We can put the above expression in equation (1) and we can get the equation 10.239

$$\frac{dP}{d\Omega} = \frac{1}{(4\pi)^2 c} 4 I^2 \frac{\cos^2\left(\omega\left(t - \frac{r}{c}\right)\right)}{\sin^4(\theta)}\left[\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right]^2 \sin^2(\theta)$$

We can arrange this like so

Now if $kl << 1$, then we can expand the $\cos(\theta) = 1 - \frac{\theta^2}{2}$

$$\frac{dP}{d\Omega} \sim \frac{I^2 \cos^2\left(\omega\left(t - \frac{r}{c}\right)\right)}{(4\pi)^2 c} \frac{\left[1 - \frac{1}{2}\left(\frac{kl}{2}\cos(\theta)\right)^2 - \left(1 - \frac{1}{2}\left(\frac{kl}{2}\right)^2\right)\right]^2}{\sin^2(\theta)} = \frac{I^2 \cos^2(\omega(t)}{(4\pi)^2 c} \frac{\left(\frac{kl}{2}\right)^4 \left[1 - \cos^2(\theta)\right]^2}{\sin^2(\theta)} = \frac{I^2 \cos^2(\omega(t)}{(4\pi)^2 c}\left(\frac{kl}{2}\right)^4 \sin^2(\theta) \quad \textbf{-- -- (A)}$$

For average we have take the average value of $\cos(\omega t)$ and by which we will come to the result

$$\frac{(d\bar{P})}{d\Omega} = \frac{(I^2)}{2c(4\pi)^2}\left(\frac{kl}{2}\right)^4 \sin^2(\theta)$$

Equation 10.207 Which tell us about the power radiated by diple

$$\frac{dP}{d\Omega} = \frac{1}{(4\pi)^2}\frac{1}{c^3}\ddot{d}^2 \sin^2(\theta) \quad ------(2)$$

Defination of the d is

$$\vec{d} = \int \vec{r}\, \rho\left(\vec{r}, t\right) d^3 r = \text{electric dipole moment}$$

$$\left(\ddot{\vec{d}}\right) = \frac{\partial^2}{\partial t^2}\int \vec{r}\, \rho\left(\vec{r}, t\right) d^3 r = \frac{\partial}{\partial t}\int j\left(\vec{r}, t\right) d^3 r = \frac{\partial}{\partial t}\sin(\omega t)\int_{-\frac{l}{2}}^{\frac{l}{2}} I \sin\left(\frac{kl}{2} - k\left|z\right|\right) dz$$

$$\left(\ddot{\vec{d}}\right) = \left(2\frac{\partial}{\partial t}\sin(\omega t)\, I\left(1 - \cos\left(\frac{kl}{2}\right)\right)\right)^2$$

Simplifying we will get

$$\left(\ddot{\vec{d}}\right)^2$$

$$\left(d\right) \;=\; 16\, c^2\, l^2 \cos^2(\omega t) \sin^4\left(\frac{kl}{4}\right)$$

Putting this result in the (2) we will get

$$\frac{dP}{d\Omega} \;=\; \frac{1}{(4\,\pi)^2}\,\frac{1}{c^3}\,16\, c^2\, l^2 \cos^2(\omega t) \sin^4\left(\frac{kl}{4}\right) \sin^2(\theta)$$

$$\frac{dP}{d\Omega} \;=\; \frac{\left(16\, l^2 \cos^2(\omega t)\right)}{(4\,\pi)^2\, c} \sin^4\left(\frac{kl}{4}\right) \sin^2(\theta)$$

Now we can expand the sine using $kl << 1$:

$$\frac{dP}{d\Omega} \;=\; \frac{\left(l^2 \cos^2(\omega t)\right)}{(4\,\pi)^2\, c} \left(\frac{kl}{2}\right)^4 \sin^2(\theta) \quad \text{------}\,(\boldsymbol{B})$$

$$\frac{\overline{(dP)}}{d\Omega} \;=\; \frac{\left(l^2\, \overline{\cos^2(\omega t)}\right)}{(4\,\pi)^2\, c} \left(\frac{kl}{2}\right)^4 \sin^2(\theta) \;=\; \frac{\left(l^2\right)}{2\, c (4\,\pi)^2} \left(\frac{kl}{2}\right)^4 \sin^2(\theta)$$

In both case we found that same power distribution . This happen because in the far field radiation limit, angular difference are negligible and dipole term is dominant so the dipole approximation makes *a* good model.

part 2

Power radiated by the half wave antenna

$$\frac{\overline{dP}}{d\Omega} \;=\; \frac{\left(2\, l^2\right)}{(4\,\pi)^2\, c}\, \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}$$

Total power radiated is given by

$$\int \frac{\overline{dP}}{d\Omega}\, d\Omega \;=\; \int \frac{\left(2\, l^2\right)}{(4\,\pi)^2\, c}\, \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}\, d\Omega$$

$$P_{hw} \;=\; \frac{\left(2\, l^2\right)}{(4\,\pi)^2\, c} \int \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}\, d\Omega$$

$$P_{hw} \;=\; \frac{\left(2\, l^2\right)}{(4\,\pi)^2\, c} \int_0^{2\,\pi} d\phi \int_0^{\pi} \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)} \sin(\theta)\, d\theta$$

Integrating

$$P_d \;= 1.22\, \frac{l^2}{4\,\pi\, c}$$

For dipole soution

$$P_d \;=\; \frac{l^2}{(4\,\pi)^2\, 2\, c} \left(\frac{kl}{2}\right)^4 2\,\pi \int_0^{\pi} \sin^3\theta\, d\theta$$

Inetgrating we will get

$$P_d \;=\; \frac{\pi^3}{48}\, \frac{l^2}{4\,\pi c}$$

$$\frac{P_{Hw}}{P_D} \;=\; 1.89$$

Hence we can say that half wave' *s* antena power is 1.89 times of the dipole approimation.

Question 2

1)

$$J_z = I\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z\,\right|\right)\delta(y)\sin(\omega \mathrm{t})[\delta(x) + \delta(x-a)]$$

$$\frac{\mathrm{dP}}{\mathrm{d}\Omega} = \frac{1}{4\pi}\,\frac{1}{c^3}\left[\hat{r} \times \int d^3 r\,\frac{\partial \vec{j}(r', t_r)}{\partial t}\right]^2$$

$$\frac{\partial \vec{j}}{\partial t} = \frac{\partial}{\partial t}\left(I\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\delta(y)\sin\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)[\delta(x) + \delta(x-a)]\right)\right)$$

$$\frac{\partial \vec{j}}{\partial t} = \left(\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\delta(y)\cos\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)[\delta(x) + \delta(x-a)]\right)\right)$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{-\frac{l}{2}}^{\frac{l}{2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\delta(x)\,\delta(y)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dx\,dy\,dz} +$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}}\int_{-\infty}^{\infty}\int_{\infty}^{\infty}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\delta(x-a)\,\delta(y)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dx\,dy\,dz}$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dz}\quad \mathrm{at}\,(x' = 0,\ y' = 0) +$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dz}\ \mathrm{at}\,((x' = a),\ y' = 0)$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dz}\quad \mathrm{at}\,(x' = 0,\ y' = 0) +$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dz}\ \mathrm{at}\,((x' = a),\ y' = 0)$$

For the first integral $\hat{r}.\vec{r}' = z'\cos(\theta)$

$$\hat{r}.\vec{r}' = (\sin(\theta)\cos(\phi),\ \sin(\theta)\sin(\phi),\ \cos(\theta))\begin{pmatrix} a \\ 0 \\ z' \end{pmatrix} = \mathrm{a}\sin(\theta)\cos(\phi) + z'\cos(\theta)$$

Let us take the first integral

$\Rightarrow$ Hence first integral is =

$$\int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)\mathrm{dz}\quad \mathrm{at}\,(x' = 0,\ y' = 0) =$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{z'\cos(\theta)}{c}\right)\right)\right)\mathrm{dz}$$

$$\Rightarrow \int_{-\frac{l}{2}}^{\frac{l}{2}}\omega \mathrm{I}\,\sin\left(\frac{\mathrm{kl}}{2} - k\,\left|\,z'\,\right|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{z'\cos(\theta)}{c}\right)\right)\right)\mathrm{dz} =$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \left\{\sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos\left(\left(\omega\left(t-\frac{r}{c}\right)\right)\cos\left(\frac{\omega z'\cos(\theta)}{c}\right) - \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\sin\left(\left(\omega\left(t-\frac{r}{c}\right)\right)\sin\left(\frac{\omega z'\cos(\theta)}{c}\right)\right)\right\}$$

dz

$$\Rightarrow \text{Ist integral} = \int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos\left(\omega\left(t-\frac{r}{c}\right)\right)\cos(k z'\cos(\theta))\,dz$$

Let us now evaluate the second integral

$$\int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos\left(\left(\omega\left(t-\frac{r}{c}+\frac{\left(\hat{r}.\vec{r'}\right)}{c}\right)\right)\right)dz \text{ at } ((x' = a),\ y' = 0) =$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi) + k z'\cos(\theta)\right)\right)dz$$

$$\Rightarrow \text{2 nd Integral} = \int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right) + k z'\cos(\theta)\right)dz'$$

In this integral also sine term will also go away and hence clearing up the clutter

$$\Rightarrow \text{2 nd Integral} = \cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right)\right)\int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos(k z'\cos(\theta))$$

Hence combining the 1 st and 2 nd Integral we can

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] =$$

$$\omega I \hat{z}\left[\cos\left(\omega\left(t-\frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right)\right)\right]\int_{\frac{-l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\left|z'\right|\right)\cos(k z'\cos(\theta))$$

Solving this integral we will get

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = 2\, I c \hat{z}\left[\cos\left(\omega\left(t-\frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right)\right)\right]\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}}{\sin^2(\theta)}$$

$$\frac{dP}{d\Omega} = \frac{I^2}{4\pi^2 c}\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\left[\cos\left(\omega\left(t-\frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right)\right)\right]^2$$

Now to calculate the averaged angular power distribution, we need to find

$$\frac{\left(\int_0^{\frac{2\pi}{\omega}}\left[\cos\left(\omega\left(t-\frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t-\frac{r}{c}\right) + k a\sin(\theta)\cos(\phi)\right)\right)\right]^2 dt\right)}{\frac{2\pi}{\omega}} = 1 + \cos(k a\sin(\theta)\cos(\phi))$$

Finally we find that time averaged angular power distribution is

$$\frac{dP}{d\Omega} = \frac{I^2}{4\pi^2 c}\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\left(1 + \cos(k a\sin(\theta)\cos(\phi))\right)$$

$$\frac{dP}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{ka}{2}\sin(\theta)\cos(\phi)\right)$$

2)

Assume that both the antenna are half wave. So $kl = \pi$

From the above results.

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi a}{2l}\sin(\theta)\cos(\phi)\right)$$

we have to now look for the time avearged angular distribution for this

Let us look when $\phi = \pm\frac{\pi}{2}$

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\cos^2\left(\frac{\pi}{2}\cos(\phi)\right)}{\sin^2(\theta)}$$ $\Rightarrow$ Constructive interference along the yz plane.

when $\theta = \frac{\pi}{2}$ and $\phi = 0,\ \pi$

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi a}{2l}\right)$$

Destructive interference will happen when $a\sin(\theta)\cos(\phi) = l$

as $\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi}{2}\right) = 0$

Question 3

Power radiated by particle of charge ze and mass m passes through a force field $\vec{F}$

$$P = \frac{((ze)^2 \gamma^2)}{6\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right] \text{ where } \vec{\beta} = \frac{\vec{v}}{c}$$

From equation 13.11 we can say that

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^3\left[(\dot{v})^2 - \frac{1}{c^2}(v \times \dot{v})^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^3\left[(a)^2 - (\boldsymbol{\beta} \times \boldsymbol{a})^2\right]$$

We have to now to express the accelration in terms of the force

$$F^\mu = m\frac{\mathrm{du}^\mu}{\mathrm{d}\tau} = m\frac{d}{\mathrm{dt}}\gamma\begin{pmatrix} c \\ v^i \end{pmatrix} = \mathrm{m}\gamma\left[\frac{\left(\vec{v}.\vec{a}\right)}{c^2}\gamma^3\begin{pmatrix} c \\ v^i \end{pmatrix} + \gamma\begin{pmatrix} 0 \\ a^i \end{pmatrix}\right]$$

Extracting 3 Vector of the force is

$$\vec{F} = \gamma^4\frac{m}{c^2}\left(\vec{v}.\vec{a}\right)\vec{v} + \mathrm{m}\gamma^2\vec{a}$$

We have to decompose the F in the parallel and perpendicular components

$$\vec{F} = \gamma^4\frac{m}{c^2}\left(\vec{v}.(a_{\|} + a_{\perp})\right)\vec{v} + \mathrm{m}\gamma^2(a_{\|} + a_{\perp})$$

$$F_{\|} + F_{\perp} = \gamma^4\frac{m}{c^2}(a_{\|})v^2 + \mathrm{m}\gamma^2(a_{\|} + a_{\perp})$$

$$F_{\|} + F_{\perp} = \gamma^4\frac{m}{c^2}(a_{\|})v^2 + \mathrm{m}\gamma^2(a_{\|}) + \mathrm{m}\gamma^2(a_{\perp})$$

$$F_{\|} = \left(\gamma^4\frac{m}{c^2}v^2 + \mathrm{m}\gamma^2\right)a_{\|} \quad \text{and } F_{\perp} = \mathrm{m}\gamma^2(a_{\perp})$$

Let us solve for the $a_{\|}$

$$F_{\|} = \mathrm{m}\gamma^2\left(\gamma^2\frac{1}{c^2}v^2 + 1\right)a_{\|} \qquad \text{writing} \quad \gamma^2\frac{1}{c^2}v^2 + 1 = \frac{1}{\left(1-\frac{v^2}{c^2}\right)}\frac{v^2}{c^2} + 1 = \frac{v^2}{c^2-v^2} + 1 = \frac{c^2}{c^2-v^2} = \gamma^2$$

$$F_{\|} = \mathrm{m}\gamma^4 a_{\|} \quad \Rightarrow a_{\|} = \frac{F_{\|}}{\mathrm{m}\gamma^4} \text{ and } a_{\perp} = \frac{F_{\perp}}{\mathrm{m}\gamma^2}$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + (a_{\perp})^2 - (\boldsymbol{\beta} \times (\boldsymbol{a}_{\|} + a_{\perp}))^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + (a_{\perp})^2 - (\boldsymbol{\beta}\,\boldsymbol{a}_{\perp})^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + \frac{(a_{\perp})^2}{\gamma^2}\right]$$

Putting the above expressions

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[\left(\frac{F_{\|}}{\mathrm{m}\gamma^4}\right)^2 + \frac{1}{\gamma^2}\left(\frac{F_{\perp}}{\mathrm{m}\gamma^2}\right)^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[\frac{F_{\|}}{m^2\gamma^8} + \frac{F_{\perp}}{m^2\gamma^6}\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[\frac{F_{\|}}{\gamma^2} + F_{\perp}\right]$$

$F^2 - \left(\vec{\beta}.\vec{F}\right)^2 = (F_{\|})^2 + (F_{\perp})^2 - \beta^2 {F_{\|}}^2 = \frac{{F_{\|}}^2}{\gamma^2} + {F_{\perp}}^2$, as well as the given charge $q = ze$ gives

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right]$$

Question 4

We have to show that relativistic result for the instantaneous energy radiated per unit time by the particle of charge ze in $\vec{E}$ and $\vec{B}$ is given by

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[\left(\vec{E} + \frac{\vec{v}}{c}\times\vec{B}\right)^2 - \left(\frac{\vec{v}}{c}.\vec{E}\right)^2\right]$$

We have found from the previous result that power emitted by the charge partice in the force field is

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right] \qquad ---------- (A)$$

Totale electromagnetic force on the particle is given by

$$\vec{F} = ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)$$

Let us calculate the

$$F^2 = ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right).ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right) = (ze)^2\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2$$

let us evaluate now

$$\vec{\beta}.\vec{F} = ze\,\vec{\beta}.\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right) = ze\left(\vec{\beta}.\vec{E} + \vec{\beta}.\frac{\vec{v}\times\vec{B}}{c}\right) = ze\left(\vec{\beta}.\vec{E}\right)$$

$$\vec{\beta}.\vec{F} = ze\left(\frac{\vec{v}}{c}.\vec{E}\right)$$

Putting everything in the (A) we will get

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[(ze)^2\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2 - \left(ze\left(\frac{\vec{v}}{c}.\vec{E}\right)\right)^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^4}{4\pi m^2 c^3}\left[\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2 - \left(\frac{\vec{v}}{c}.\vec{E}\right)^2\right]$$

Question 5

a

Energy radiated into the solid angle dΩ per unit retarded time is given by

Equation 15.25 for the 1 particle is written by

$$\frac{dP(t')}{d\Omega} = \frac{\frac{e^2}{(4\pi)^2 c^3}\left[\hat{r}\times\left(\vec{a}+\hat{r}\times\left(\frac{\vec{v}}{c}\times\vec{a}\right)\right)\right]^2}{\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} \text{ ---- } (B)$$

$$z(t') = a\cos(\omega_o t')$$

$$\dot{z}(t') = v_z(t') = -a\omega_o \sin(\omega_o t')$$

$$\ddot{z}(t') = a_z(t') = -a(\omega_o)^2\cos(\omega_o t')$$

$$\dot{z}(t') \times \ddot{z}(t') = \vec{v}\times\vec{a} = 0$$

Hence the equation B can be writtten as a

$$\frac{dP(t')}{d\Omega} = \frac{e^2\left[\hat{r}\times\vec{a}\right]^2}{(4\pi)^2 c^3\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} \text{ ---------- } C$$

let us calculate the $\hat{r}\times\vec{a}$ and $\vec{v}.\hat{r}$ for our case

$$\hat{r}\times\vec{a} = a(\omega_o)^2\cos(\omega_o t')\left(\sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}\right)\times\hat{z}$$

$$\hat{r}\times\vec{a} = a(\omega_o)^2\cos(\omega_o t')\left(-\sin(\theta)\cos(\phi)\,\hat{y} + \sin(\theta)\sin(\phi)\,\hat{x}\right)$$

$$\left(\hat{r}\times\vec{a}\right)^2 = a^2(\omega_o)^4\cos^2(\omega_o t')\,\sin^2(\theta)$$

$$1-\frac{\vec{v}}{c}.\hat{r} = 1 + \frac{1}{c}\left(a\omega_o\sin(\omega_o t')\,\hat{z}\,.\left(\sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}\right)\right)$$

$$1-\frac{\vec{v}}{c}.\hat{r} = 1 + \frac{1}{c}\left(a\omega_o\cos(\theta)\sin(\omega_o t')\right) = 1 + \beta\cos(\theta)\sin(\omega_o t')$$

Putting everyting on the equation C

$$\frac{dP(t')}{d\Omega} = \frac{e^2\left[\hat{r}\times\vec{a}\right]^2}{(4\pi)^2 c^3\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} = \frac{e^2\left(a^2(\omega_o)^4\cos^2(\omega_o t')\,\sin^2(\theta)\right)}{(4\pi)^2 c^3(1+\beta\cos(\theta)\sin(\omega_o t'))^5}$$

which can be further written as a

$$\frac{dP(t')}{d\Omega} = \frac{e^2 c\,\beta^4\cos^2(\omega_o t')\,\sin^2(\theta)\,)}{(4\pi)^2 a^2(1+\beta\cos(\theta)\sin(\omega_o t'))^5}$$

b

$$\frac{\overline{dP}(t')}{d\Omega} = \int\frac{dP(t')}{d\Omega}\,dt' = \frac{\left(\int_0^{\frac{(2\pi)}{\omega}}\frac{e^2 c\,\beta^4\cos^2(\omega_o t')\,\sin^2(\theta)}{(4\pi)^2 a^2(1+\beta\cos(\theta)\sin(\omega_o t'))^5}\,dt'\right)}{\frac{2\pi}{\omega_0}}$$

$$\frac{\overline{dP}(t')}{d\Omega} = \frac{\omega_0\left(e^2 c\,\beta^4\sin^2(\theta)\right)}{2\pi(4\pi)^2 a^2}\int_0^{\frac{(2\pi)}{\omega_0}}\frac{\cos^2(\omega_o t')\;)}{(1+\beta\cos(\theta)\sin(\omega_o t'))^5}\,dt'$$

Calcuating the inside the integration in the mathematica

$$\frac{\overline{dP}(t')}{d\Omega} = \frac{e^2 c\,\beta^4\sin^2(\theta)}{128\pi)^2 a^2}\,\frac{(4+\beta^2\cos^2(\theta))}{(1-\beta^2\cos^2(\theta))^{\frac{7}{2}}}$$

Question 6

We want to calculate and sketch the frequency distribution for a particle which uniformly decelerates over some time interval τ

accelration $a = \left\{ \begin{array}{ll} \frac{(v_f - v_i)}{\tau}, & \frac{-\tau}{2} \leq t \leq \frac{\tau}{2} \\ 0, & \text{otherwise} \end{array} \right\}$

Let us fourier transform of the accelaration

$$\vec{a}(\omega) = \frac{\hat{2}}{2\pi} \int_{\frac{-\tau}{2}}^{\frac{\tau}{2}} \frac{(v_f - v_i)}{\tau} e^{i\omega t} dt = \frac{1}{i\omega} \frac{(v_f - v_i)}{2\pi\tau} \left(e^{\frac{i\omega\tau}{2}} - e^{-\frac{i\omega\tau}{2}} \right) = \frac{1}{\omega} \frac{(v_f - v_i)}{\pi\tau} \sin\left(\frac{\omega\tau}{2}\right)$$

Energy distribution per unit solid angle and frequncy is given by

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} \left[\hat{r} \times \vec{a}(\omega) \right]^2$$

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} (a(\omega))^2 \sin^2(\theta)$$

$$\frac{dE}{d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} (a(\omega))^2 \int d\Omega \sin^2(\theta)$$

putting this into the mathematica we will get

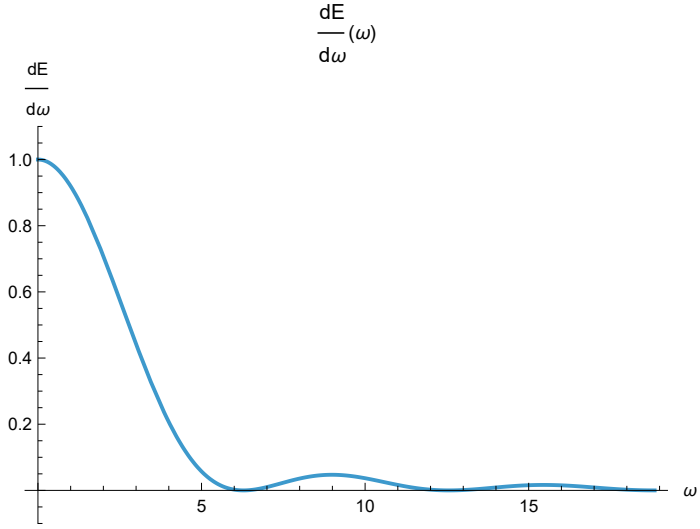
$$\frac{dE}{d\omega} = \frac{e^2}{6\pi^4 c^3} \left(\frac{(v_f - v_i)}{\tau} \right)^2 \frac{\sin^2\left(\frac{\omega\tau}{2}\right)}{\omega^2}$$

$$\frac{dE}{d\omega} = \frac{e^2}{24\pi^4 c^3} ((v_f - v_i))^2 \frac{\sin^2\left(\frac{\omega\tau}{2}\right)}{\left(\frac{\omega\tau}{2}\right)^2}$$

variation of $\frac{dE}{d\omega}$ will be like

In[5]:=
```
Plot[Sin[x/2]^2/(x/2)^2, {x, 0, 6 π}, PlotRange → {-0.1, 1.1},
 PlotLabel → Style["dE/dω (ω)"], AxesLabel → {"ω", "dE/dω"}]
```

Out[5]=



Question 7

A charge e of mass m is going through electric field $(E_o, 0, 0)$ with velocity $(v_0, 0, 0)$ for a distcane x_0
Accelrationn experienced by the charged particle is

$a(t) = \frac{eE_0}{m}$ for $0 < t < t_f$

let us find the fourier transform of the accelaration

$\tilde{a}(\omega) = \frac{1}{2\pi}\int_0^{t_f} a(t)\, e^{i\omega t}\, dl\, \omega$

$\tilde{a}(\omega) = \frac{eE_0}{2\pi i m\omega}\left(e^{i\omega t_f} - 1\right)$

$\tilde{a}(\omega) = \frac{(e E_0)}{\pi m\omega}\sin\left(\frac{\omega t_f}{2}\right) e^{\frac{i\omega t_f}{2}}$

let us now find out the time using the kinematic equation of motion

$x = ut - \frac{1}{2} a t^2$

$\frac{mx_o}{eE_o} = \frac{mv_o}{eE_o} t_f - \frac{1}{2}\frac{eE_0}{m}(t_f)^2$

$\frac{mx_o}{eE_o} + \frac{m^2(v_o)^2}{4e^2(E_o)^2} = \left(t_f + \frac{mv_0}{2eE_o}\right)^2$

hence $t_f = \sqrt{\frac{m}{eE_o}\left(x_o + \frac{mv_o}{4eE_o}\right)} - \frac{mv_o}{2eE_o}$

$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}(a(\omega))^2 \sin^2(\theta)$

$\frac{dE}{d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}(a(\omega))^2 \int d\Omega \sin^2(\theta)$

Putting the value of $a(\omega)$ and calcuting the integral again like in previous question we will get

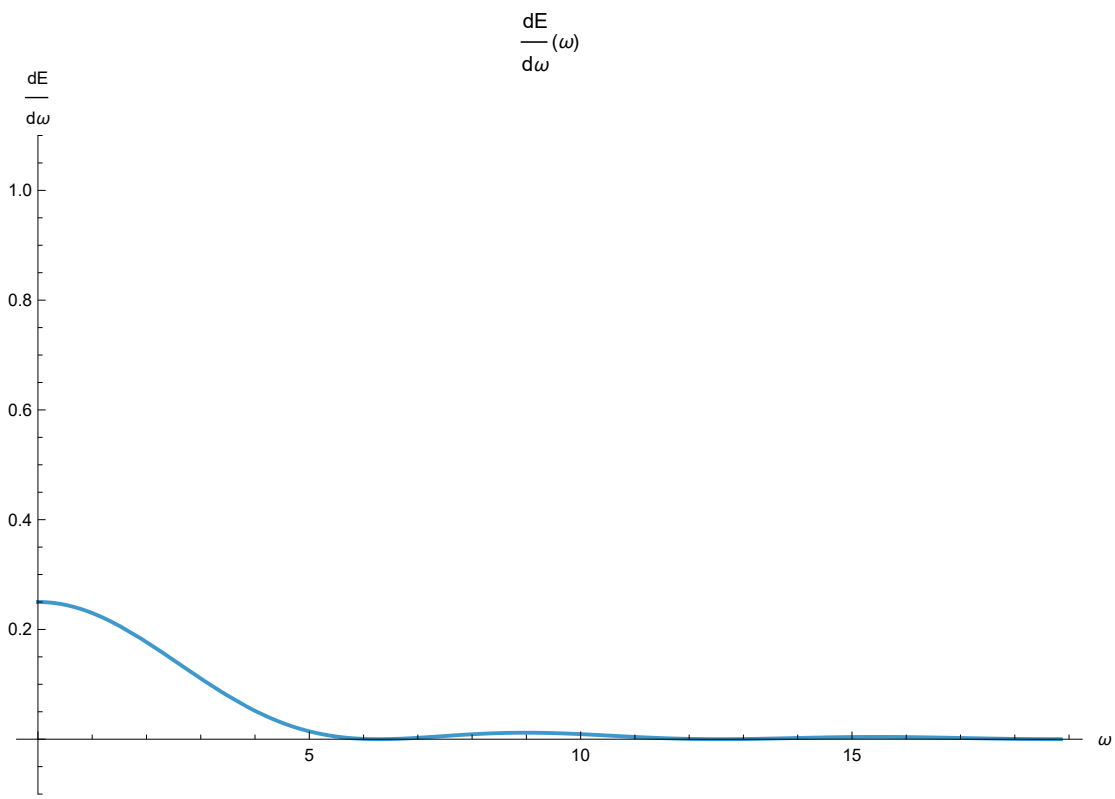
$\frac{dE}{d\omega} = \left(\frac{(e^2 E_o)}{4\pi^2 m c}\right)^2 \frac{2}{c\omega^2}\sin^2\left(\frac{\omega t_f}{2}\right)\frac{4}{3}$

$\frac{dE}{d\omega} = \frac{e^4 (t_f E_o)^2}{6\pi^4 m c^3}\frac{\sin^2\left(\frac{\omega t_f}{2}\right)}{(t_f\omega)^2}\frac{4}{3}$

Plot will be almost same like above

In[8]:=
```
Plot[Sin[x/2]^2/(x)^2, {x, 0, 6 π}, PlotRange → {-0.1, 1.1},
 PlotLabel → Style["dE/dω (ω)"], AxesLabel → {"ω", "dE/dω"}, ImageSize → Large]
```

Out[8]=



Question 8

Given

v(t) = $\vec{v}_o$ = constant

$\dot{v}(t) = 0$

$\vec{v}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \vec{v}_o\, e^{i\omega t}\, dt = \vec{v}_o\, \delta(\omega)$

$\vec{a}(\omega) = 0$

Plugging into the expression that are given in the questions

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}\left[\hat{r}\times\vec{v}(\omega)\right]^2 = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}\left[\hat{r}\times\vec{v}_o\,\delta(\omega)\right]^2 = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}(v_o)^2\sin^2(\theta)\,\delta^2(\omega)$$

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}\left[\hat{r}\times\vec{a}(\omega)\right]^2 = 0$$

Question 9

Photon density per unit lenght

$\frac{dN}{dz} = \frac{1}{137\,c} \int_{\omega_2}^{\omega_1} 1 - \cos^2(\theta_c)\, d\omega$ ------**D**

where $\mathrm{Cos}(\theta_c) = \frac{1}{\beta n} = \frac{c}{vn}$

we have to find the θ_c , ω_1, ω_2

$\lambda_1 = 4000\,\dot{A}$, $\omega_1 = \frac{(2\pi c)}{\lambda_1} = 4.712 \times 10^{15}$

$\lambda_2 = 6000\,\dot{A}$, $\omega_1 = \frac{(2\pi c)}{\lambda_1} = 3.142 \times 10^{15}$

refractive index = 1.5

KE (Proton) = 10^6 eV = $1.602 \times 10^{-13}\,J$

KE (Proton) = 10^6 eV = $1.602 \times 10^{-10}\,J$

$m(p) = 1.6 \times 10^{-27}$ kg

$m(e) = 9.109 \times 10^{-31}$ kg

$v(e) = \sqrt{2\,\frac{k}{m}} = 4.736 \times 10^8\, m/s$

$v(p) = \sqrt{2\,\frac{k}{m}} = 5.931 \times 10^8\, m/s$

$\theta_c(e) = 70.2$ degree $\theta_c(p) = 62.8$ degree

Putting everything into D and finding

$\frac{dN}{dz}$ for the proton = $30\,236\, m^{-1} \sim 302.36\ \mathrm{cm}^{-1}$

$\frac{dN}{dz}$ for the electron = $33\,873\, m^{-1} \sim 338\ \mathrm{cm}^{-1}$

Hence cherenkov angle for the electron and proton are 70.2 and 62.8 degree respectively. And number of photon emitted are 339 and 302 respectively.

Energy radiated into the solid angle dΩ per unit retarded time is given by

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi}\frac{\left(\left|\vec{n}\times\left[\left(\vec{n}-\vec{v}\right)\times\vec{a}\right]\right|^2\right)}{\left(1-\vec{n}.\vec{v}\right)^5} \quad ----(1)$$

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi}\frac{\left(\left|\vec{n}\times\left[\left(\vec{n}-\vec{v}\right)\times\vec{a}\right]\right|^2\right)}{\left(1-\vec{n}.\vec{v}\right)^5}$$

$$z(t') = a\cos(\omega_o t')$$

$$\dot{z}(t') = v_z(t') = -a\omega_o \sin(\omega_o t')$$

$$\ddot{z}(t') = a_z(t') = -a(\omega_o)^2\cos(\omega_o t')$$

$$\dot{z}(t') \times \ddot{z}(t') = \vec{v}\times\vec{a} = 0$$

Plugging in these resylts to the power distribution gives

$$\vec{n} = \sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}$$

$$\left(\vec{n}-\vec{v}\right) = \sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + (\cos(\theta) - v_z)\,\hat{z}$$

$$\left(\vec{n}-\vec{v}\right)\times(a_z)\,\hat{z} = (a_z)\sin(\theta)\cos(\phi)\left(-\hat{y}\right) + a_z\sin(\theta)\sin(\phi)\,\hat{x}$$

$$\left(\vec{n}-\vec{v}\right)\times a_z\,\hat{z} = a_z\sin(\theta)\left(-\cos(\phi)\,\hat{y} + \sin(\phi)\,\hat{x}\right)$$

Let us compute

$$\vec{n}\times\left[\left(\vec{n}-\vec{v}\right)\times\vec{a}\right] = \left(\sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}\right)\times a_z\sin(\theta)\left(-\cos(\phi)\,\hat{y} + \sin(\phi)\,\hat{x}\right)$$

$$\vec{n}\times\left[\left(\vec{n}-\vec{v}\right)\times\vec{a}\right] = -a_z\sin(\theta)\cos(\phi)\sin(\theta)\cos(\phi)\,\hat{z} +$$
$$a_z\sin(\theta)\cos(\theta)\cos(\phi)\,\hat{x} - a_z\sin(\theta)\cos(\phi)\sin(\theta)\cos(\phi)\,\hat{z} + \cos(\theta)\,a_z\sin(\theta)\sin(\phi)\,\hat{y}$$

$$\vec{n}\times\left[\left(\vec{n}-\vec{v}\right)\times\vec{a}\right] =$$
$$-a_z\sin^2(\theta)\cos^2(\phi)\,\hat{z} + a_z\sin(\theta)\cos(\theta)\cos(\phi)\,\hat{x} - a_z\sin^2(\theta)\cos^2(\phi)\,\hat{z} + \cos(\theta)\,a_z\sin(\theta)\sin(\phi)\,\hat{y}$$